

Section 9.1 - Question 31

Let A be the set of students at your school and B the set of books in the school library. Let R_1 and R_2 be the relations consisting of all ordered pairs (a, b) , where student a is required to read book b in a course, and where student a has read book b , respectively. Describe the ordered pairs in each of these relations.

a) $R_1 \cup R_2$ The ordered pairs (a, b) in the union of R_1 and R_2 are those where student a is either required to read book b from the school library in a course and/or has read book b .

b) $R_1 \cap R_2$ The ordered pairs (a, b) in the intersection of R_1 and R_2 describe instances in which a student a is required to read book b in a course and has read book b .

c) $R_1 \oplus R_2$ The ordered pairs (a, b) in the symmetric difference of R_1 and R_2 are those where student a is required to read book b in a course or has read book b , but not both.

d) $R_1 - R_2$ The ordered pairs (a, b) in the difference of R_1 and R_2 are those where student a is required to read book b in a course, but has not read book b .

e) $R_2 - R_1$ The ordered pairs (a, b) in the difference of R_2 and R_1 are those where student a has read book b , but was not required to read book b in a course.

Section 9.1 - Question 33

Let R be the relation on the set of people consisting of pairs (a, b) , where a is a parent of b . Let S be the relation on the set of people consisting of pairs (a, b) , where a and b are siblings (brothers or sisters). What are $S \circ R$ and $R \circ S$?

$S \circ R$: The composition of S and R is the relation on the set of people consisting of pairs (a, b) , where a is a parent of b and a and b are siblings. This relation would describe all parent-child relationships in which the parent has multiple children or the parent's child has half- or step-siblings.

$R \circ S$: The composition of R and S is the relation on the set of people consisting of pairs (a, b) , where a and b are siblings and a is a parent of b . This relation would describe relationships between aunts/uncles and nieces/nephews.

Section 9.1 - Question 34h

$R_1 = \{(a, b) \in R^2 | a > b\}$, the greater than relation, $R_2 = \{(a, b) \in R^2 | a \geq b\}$, the greater than or equal to relation,

$R_3 = \{(a, b) \in R^2 | a < b\}$, the less than relation, $R_4 = \{(a, b) \in R^2 | a \leq b\}$, the less than or equal to relation,

$R_5 = \{(a, b) \in R^2 | a = b\}$, the equal to relation, $R_6 = \{(a, b) \in R^2 | a \neq b\}$, the unequal to relation.

Find $R^2 \oplus R_4$

$$R_2 \oplus R_4 = \{(a, b) \in R^2 | a \geq b\} \oplus \{(a, b) \in R^2 | a \leq b\} = \{(a, b) \in R^2 | a \neq b\} = R_6$$

Section 9.1 - Question 36a

$R_1 = \{(a, b) \in R^2 | a > b\}$, the greater than relation, $R_2 = \{(a, b) \in R^2 | a \geq b\}$, the greater than or equal to relation,

$R_3 = \{(a, b) \in R^2 | a < b\}$, the less than relation, $R_4 = \{(a, b) \in R^2 | a \leq b\}$, the less than or equal to relation,

$R_5 = \{(a, b) \in R^2 | a = b\}$, the equal to relation, $R_6 = \{(a, b) \in R^2 | a \neq b\}$, the unequal to relation.

Find $R_1 \circ R_1$

$$R_1 \circ R_1 = \{(a, b) \in R^2 | a > b\} \circ \{(a, b) \in R^2 | a > b\} = \{(a, b) \in R^2 | a > b\} = R_1$$

Section 9.1 - Question 37

a) $R_2 \circ R_1$ $R_2 \circ R_1 = \{(a, b) \in R^2 | a \geq b\} \circ \{(a, b) \in R^2 | a > b\} = \{(a, b) \in R^2 | a > b\} = R_1$

b) $R_2 \circ R_2$ $R_2 \circ R_2 = \{(a, b) \in R^2 | a \geq b\} \circ \{(a, b) \in R^2 | a \geq b\} = \{(a, b) \in R^2 | a \geq b\} = R_2$

c) $R_3 \circ R_5$ $R_3 \circ R_5 = \{(a, b) \in R^2 | a < b\} \circ \{(a, b) \in R^2 | a = b\} = \{(a, b) \in R^2 | a < b\} = R_3$

d) $R_4 \circ R_1$ $R_4 \circ R_1 = \{(a, b) \in R^2 | a \leq b\} \circ \{(a, b) \in R^2 | a > b\} = R^2$

e) $R_5 \circ R_3$ $R_5 \circ R_3 = \{(a, b) \in R^2 | a = b\} \circ \{(a, b) \in R^2 | a < b\} = \{(a, b) \in R^2 | a < b\} = R_3$

f) $R_3 \circ R_6$ $R_3 \circ R_6 = \{(a, b) \in R^2 | a < b\} \circ \{(a, b) \in R^2 | a \neq b\} = R^2$

g) $R_4 \circ R_6$ $R_4 \circ R_6 = \{(a, b) \in R^2 | a \leq b\} \circ \{(a, b) \in R^2 | a \neq b\} = R^2$

h) $R_6 \circ R_6$ $R_6 \circ R_6 = \{(a, b) \in R^2 | a \neq b\} \circ \{(a, b) \in R^2 | a \neq b\} = R^2$

Section 9.1 - Question 39

Find the relations S^2 for $i = 1, 2, 3, 4, 5, 6$ where

$S_1 = \{(a, b) \in Z^2 | a > b\}$, the greater than relation,

$$S_1^2 = \{(a, b) \in Z^2 | a > b\} \circ \{(a, b) \in Z^2 | a > b\} = \{(a, b) \in Z^2 | a - 1 > b\}$$

$S_2 = \{(a, b) \in Z^2 | a \geq b\}$, the greater than or equal to relation,

$$S_2^2 = \{(a, b) \in Z^2 | a \geq b\} \circ \{(a, b) \in Z^2 | a \geq b\} = \{(a, b) \in Z^2 | a \geq b\}$$

$S_3 = \{(a, b) \in Z^2 | a < b\}$, the less than relation,

$$S_3^2 = \{(a, b) \in Z^2 | a < b\} \circ \{(a, b) \in Z^2 | a < b\} = \{(a, b) \in Z^2 | a < b - 1\}$$

$S_4 = \{(a, b) \in Z^2 | a \leq b\}$, the less than or equal to relation,

$$S_4^2 = \{(a, b) \in Z^2 | a \leq b\} \circ \{(a, b) \in Z^2 | a \leq b\} = \{(a, b) \in Z^2 | a \leq b\}$$

$S_5 = \{(a, b) \in Z^2 | a = b\}$, the equal to relation,

$$S_5^2 = \{(a, b) \in Z^2 | a = b\} \circ \{(a, b) \in Z^2 | a = b\} = \{(a, b) \in Z^2 | a = b\}$$

$S_6 = \{(a, b) \in Z^2 | a \neq b\}$, the unequal to relation.

$$S_6^2 = \{(a, b) \in Z^2 | a \neq b\} \circ \{(a, b) \in Z^2 | a \neq b\} = Z^2$$

Section 9.1 - Question 41

Let R be the relation on the set of people with doctorates such that $(a, b) \in R$ if and only if a was the thesis advisor of b . When is an ordered pair (a, b) in R^2 ? When is an ordered pair (a, b) in R^n , when n is a positive integer? (Assume that every person with a doctorate has a thesis advisor.)

- An ordered pair (a, b) is in R^2 if and only if a was the thesis advisor of the thesis advisor of b .
- An ordered pair (a, b) is in R^n for $n > 2$ if and only if a was the thesis advisor of the thesis advisor of the thesis advisor of ... of b , where the number of iterations is n .

Section 9.1 - Question 43

Let R_1 and R_2 be the “congruent modulo 3” and the “congruent modulo 4” relations, respectively, on the set of integers. That is, $R_1 = \{(a, b) | a \equiv b(\text{mod } 3)\}$ and $R_2 = \{(a, b) | a \equiv b(\text{mod } 4)\}$

a) $R_1 \cup R_2$ $R_1 \cup R_2 = \{(a, b) | a \equiv b(\text{mod } 3)\} \cup \{(a, b) | a \equiv b(\text{mod } 4)\} = \{(a, b) | a \equiv b(\text{mod } 3) \vee a \equiv b(\text{mod } 4)\}$

b) $R_1 \cap R_2$ $R_1 \cap R_2 = \{(a, b) | a \equiv b(\text{mod } 3)\} \cap \{(a, b) | a \equiv b(\text{mod } 4)\} = \{(a, b) | a \equiv b(\text{mod } 12)\}$

c) $R_1 - R_2$ $R_1 - R_2 = \{(a, b) | a \equiv b(\text{mod } 3)\} - \{(a, b) | a \equiv b(\text{mod } 4)\} = \{(a, b) | a \equiv b(\text{mod } 3) \wedge a \not\equiv b(\text{mod } 4)\}$

d) $R_2 - R_1$ $R_2 - R_1 = \{(a, b) | a \equiv b(\text{mod } 4)\} - \{(a, b) | a \equiv b(\text{mod } 3)\} = \{(a, b) | a \equiv b(\text{mod } 4) \wedge a \not\equiv b(\text{mod } 3)\}$

e) $R_1 \oplus R_2$ $R_1 \oplus R_2 = \{(a, b) | a \equiv b(\text{mod } 3)\} \oplus \{(a, b) | a \equiv b(\text{mod } 4)\} = \{(a, b) | a \equiv b(\text{mod } 3) \vee a \equiv b(\text{mod } 4) \wedge a \not\equiv b(\text{mod } 12)\}$

Section 9.2 - Question 1

List the triples in the relation $\{(a, b, c) \mid a, b, \text{ and } c \text{ are integers with } 0 < a < b < c < 5\}$.

- (1, 2, 3)
- (1, 2, 4)
- (1, 3, 4)
- (2, 3, 4)

Section 9.2 - Question 2

Which 4-tuples are in the relation $\{(a, b, c, d) \mid a, b, c, \text{ and } d \text{ are positive integers with } abcd = 6\}$?

- (6, 1, 1, 1)
- (1, 6, 1, 1)
- (1, 1, 6, 1)
- (1, 1, 1, 6)
- (1, 1, 2, 3)
- (1, 2, 3, 1)
- (2, 3, 1, 1)
- (1, 1, 3, 2)
- (1, 3, 2, 1)
- (3, 2, 1, 1)
- (1, 2, 1, 3)
- (2, 1, 3, 1)
- (1, 3, 1, 2)
- (3, 1, 2, 1)
- (2, 1, 1, 3)
- (3, 1, 1, 2)

Section 9.2 - Question 3

List the 5-tuples in the relation in Table 8.

<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44

Figure 1: Table 8

- (Nadir, 122, 34, Detroit, 08:10)
- (Acme, 221, 22, Denver, 08:17)
- (Acme, 122, 33, Anchorage, 08:22)
- (Acme, 323, 34, Honolulu, 08:25)
- (Nadir, 199, 13, Detroit, 08:47)
- (Acme, 222, 22, Denver, 09:10)
- (Nadir, 322, 34, Detroit, 09:44)

Section 9.2 - Question 5

<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44

Figure 1: Table 8

Composite keys containing *Airline* field: Airline and flight number, airline and departure time

Section 9.2 - Question 9

The 5-tuples in a 5-ary relation represent these attributes of all people in the United States: name, Social Security number, street address, city, state.

a)

Determine a primary key for this relation.

The Social Security number is a unique identifier for each person in the United States and can be used as a primary key for this relation.

b)

Under what conditions would (name, street address) be a composite key?

If two people in the United States have the same name and street address, then (name, street address) would not be a unique identifier and would not be a composite key. For this to be a composite key, (a) each person in the United States would have a unique name, (b) each person in the United States would have a unique street address, or (c) the combination of name and street address would be unique for each person in the United States.

c)

Under what conditions would (name, street address, city) be a composite key?

If two people in the United States have the same name, street address, and city, then (name, street address, city) would not be a unique identifier and would not be a composite key. For this to be a composite key, (a) each person in the United States would have a unique name, (b) each person in the United States would have a unique street address + city, (c) each person in the United States would have a unique city, or (d) the combination of name, street address, and city would be unique for each person in the United States.

Section 9.2 - Question 10

What do you obtain when you apply the selection operator s_C , where C is the condition Room = A100, to the database in Table 7?

Table 7

Professor	Department	Course_number	Room	Time
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.
Farber	Psychology	617	A110	11:00 A.M.
Grammer	Physics	544	B505	4:00 P.M.
Rosen	Computer Science	518	N521	2:00 P.M.
Rosen	Mathematics	575	N502	3:00 P.M.

The selection operator s_C , where C is the condition Room = A100, applied to the database in Table 7 would return the following tuples:

Professor	Department	Course_number	Room	Time
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.

Section 9.2 - Question 15

Which projection mapping is used to delete the first, second, and fourth components of a 6-tuple?

The projection mapping used to delete the first, second, and fourth components of a 6-tuple is $\pi\{3, 5, 6\}$.

Section 9.2 - Question 28b

a) What are the operations that correspond to the query expressed using this SQL statement?

```
SELECT Supplier
FROM Part_needs
WHERE 1000 ≤ Part number ≤ 5000
```

Projection on Supplier attribute from the records in the Part_needs table that satisfies the condition $1000 \leq \text{Part number} \leq 5000$.

b) What is the output of this query given the database in Table 11 as input?

Supplier	Part_number	Project
23	1092	1
23	1101	3
23	9048	4
31	4975	3
31	3477	2
32	6984	4
32	9191	2
33	1001	1

The output of this query given the database in Table 11 as input is:

Supplier
23
31
33

Section 9.2 - Question 31

Let R be the relation on $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}^+$ consisting of triples (a, b, m) , where a, b , and m are integers with $m \geq 1$ and $a \equiv b \pmod{m}$. Then $(8, 2, 3)$, $(-1, 9, 5)$, and $(14, 0, 7)$ all belong to R , but $(7, 2, 3)$, $(-2, -8, 5)$, and $(11, 0, 6)$ do not belong to R because $8 \not\equiv 2 \pmod{3}$, $-1 \not\equiv 9 \pmod{5}$, and $14 \not\equiv 0 \pmod{7}$, but $7 \equiv 2 \pmod{3}$, $-2 \equiv -8 \pmod{5}$, and $11 \equiv 0 \pmod{6}$. This relation has degree 3 and its first two domains are the set of all integers and its third domain is the set of positive integers.

Determine whether there is a primary key for this relation.

To give counterexample for each field, consider the following:

- For a
 - $8 \equiv 14 \pmod{6}$ and $8 \equiv 2 \pmod{4}$
- For b
 - $8 \equiv 2 \pmod{3}$ and $9 \equiv 2 \pmod{7}$
- For m
 - $3 \equiv 4 \pmod{1}$ and $5 \equiv 6 \pmod{1}$

Thus, no field can be used as a primary key for this relation because there is at least one counterexample for each field wherein a value is in the set of integers but is in the same equivalence class as another value in the set of integers.

Section 9.3 - Question 2c

$\{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$

Represent this relation on $\{1, 2, 3, 4\}$ with a matrix (with the elements of this set listed in increasing order).

	1	2	3	4
1	0	1	1	1
2	1	0	1	1
3	1	1	0	1
4	1	1	1	0

Section 9.3 - Question 3

List the ordered pairs in the relations on $\{1, 2, 3\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Figure 1: Matrices

- a)** $(1, 1), (1, 3), (2, 2), (3, 1), (3, 3)$
- b)** $(1, 2), (2, 2), (3, 2)$
- c)** $(1, 1), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (3, 3)$

Section 9.3 - Question 5

How can the matrix representing a relation R on a set A be used to determine whether the relation is irreflexive?

If the diagonal of the matrix representing a relation R on a set A contains all 0s, then the relation is irreflexive. This is because the diagonal of the matrix represents the ordered pairs (a, a) for each element a in the set A , and if all of these pairs are 0, then no element is related to itself, making the relation irreflexive.

Section 9.3 - Question 7

Determine whether the relations represented by the matrices in Exercise 3 are reflexive, irreflexive, symmetric, anti-symmetric, and/or transitive.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Figure 1: Matrices

- **a)** Reflexive, symmetric, transitive
- **b)** Antisymmetric, transitive
- **c)** Symmetric

Section 9.3 - Question 8

Determine whether the relations represented by the matrices in Exercise 4 are reflexive, irreflexive, symmetric, anti-symmetric, and/or transitive.

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Figure 1: Matrices from Exercise 4

- **a)** Symmetric
- **b)** Reflexive
- **c)** Irreflexive, symmetric

Section 9.3 - Question 9

How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 100\}$ consisting of the first 100 positive integers have if R is

- a)** $\{(a, b) | a > b\}$ $99 + 98 + 97 + \dots + 1$ is an arithmetic series with a common difference of -1 and a first term of 99 . The sum of the first 99 positive integers is $\frac{99(99+1)}{2} = 4950$.
- b)** $\{(a, b) | a \neq b\}$ Each number in the set $A = \{1, 2, 3, \dots, 100\}$ can be related to 99 other numbers in the set, resulting in $100 * 99 = 9900$ ordered pairs.
- c)** $\{(a, b) | a = b + 1\}$ 99 ordered pairs, as each number in the set $A = \{1, 2, 3, \dots, 100\}$ is related to the number that is one greater than it except for 100, which is not related to any number in this relation.
- d)** $\{(a, b) | a = 1\}$ 100 ordered pairs, as the number 1 is related to every number in the set $A = \{1, 2, 3, \dots, 100\}$.
- e)** $\{(a, b) | ab = 1\}$ 1 ordered pair, as the only pair of numbers in the set $A = \{1, 2, 3, \dots, 100\}$ that satisfies the condition $ab = 1$ is $(1, 1)$.

Section 9.3 - Question 10a

How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 1000\}$ consisting of the first 1000 positive integers have if R is $\{(a, b) | a \leq b\}$?

$1 + 2 + 3 + \dots + 1000$ is an arithmetic series with a common difference of 1 and a first term of 1. The sum of the first 1000 positive integers is $\frac{1000(1000+1)}{2} = 500500$.

Section 9.3 - Question 12

How can the matrix for R^{-1} , the inverse of the relation R , be found from the matrix representing R , when R is a relation on a finite set A ?

If R is represented by the matrix $m_{ij} \begin{cases} 1 & \text{if } (a_i, a_j) \in R \\ 0 & \text{if } (a_i, a_j) \notin R \end{cases}$

Then if $R^{-1} = \{(b, a) | (a, b) \in R\}$ it should be represented by the matrix $m_{ji} \begin{cases} 1 & \text{if } (a_j, a_i) \in R \\ 0 & \text{if } (a_j, a_i) \notin R \end{cases}$

So the matrix for R^{-1} can be found by transposing the matrix for R . This means that the rows of the matrix for R become the columns of the matrix for R^{-1} and vice versa.

Section 9.3 - Question 13

Let R be the relation represented by the matrix

	1	2	3
1	0	1	1
2	1	1	0
3	1	0	1

a) R^{-1}

	1	2	3
1	0	1	1
2	1	1	0
3	1	0	1

b) $\overline{R}$

	1	2	3
1	1	0	0
2	0	0	1
3	0	1	0

c) R^2

	1	2	3
1	1	1	1
2	1	1	1
3	1	1	1

Section 9.3 - Question 14

Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

a) $R_1 \cup R_2$

	1	2	3
1	0	1	0
2	1	1	1
3	1	1	1

b) $R_1 \cap R_2$

	1	2	3
1	0	1	0
2	0	1	1
3	1	0	0

c) $R_2 \circ R_1$

	1	2	3
1	1	1	1
2	1	1	1
3	1	1	1

d) $R_1 \circ R_1$

	1	2	3
1	1	1	1
2	1	1	1

	1	2	3
3	1	1	1

e) $R_1 \oplus R_2$

	1	2	3
1	0	0	0
2	1	0	0
3	1	1	1

Section 9.3 - Question 15

Let R be the relation represented by the matrix

	1	2	3
1	0	1	0
2	0	0	1
3	1	1	0

Find the matrices that represent

a) R^2

	1	2	3
1	0	0	1
2	1	1	0
3	0	1	1

b) R^3

	1	2	3
1	1	1	0
2	0	1	1
3	1	1	1

c) R^4

	1	2	3
1	0	1	1
2	1	1	1
3	1	1	1

Section 9.4 - Question 1

Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the

a) reflexive closure of R The reflexive closure of R is the union of R and the set of ordered pairs (a, a) for each element a in the set $\{0, 1, 2, 3\}$. The reflexive closure of R is:

$\{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0), (0, 0), (1, 1), (2, 2), (3, 3)\}$

b) symmetric closure of R The symmetric closure of R is the union of R and the set of ordered pairs (b, a) for each ordered pair (a, b) in R . The symmetric closure of R is:

$\{(0, 1), (1, 0), (1, 1), (1, 2), (2, 1), (2, 0), (2, 2), (3, 0), (0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (0, 0), (3, 0)\}$

Section 9.4 - Question 2

Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers.
What is the reflexive closure of R ?

$\mathbb{Z}^2$

The reflexive closure of R is the union of R and the set of ordered pairs (a, a) for each element a in the set of integers. Since the relation R is defined as $\{(a, b) \mid a \neq b\}$, the reflexive closure of R is the set of all ordered pairs of integers.